Shantanu Amit More

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Availability – August 25, 2026 to December 30, 2026

EDUCATION

Clemson University | GPA: 4.0/4.0
Master of Science in Automotive Engineering — August 2025 – Present
Relevant courses – Advanced and Electric Powertrain, Automotive Electronics Integration, Safety and Stability Systems

College of Engineering, Pune (COEP) | GPA: 8.1/10
Bachelor of Technology, Production Engineering and Industrial Management — August 2019 - July 2023

EXPERIENCE

Deep Orange 18, Clemson University — January 2026 – May 2026
- Modeled and sized the powertrain for an autonomous NASCAR-style vehicle with the required top speed of 200 mph.

Clemson University - *Research Assistant (Prof. Jianfeng Zhang)* — August 2025 – May 2026
- Simulated HPPC test on AVL software. Formulation of the SoC estimation of cell using simple Kalman filter in MATLAB.

Binani Technology India - *Mechanical Design Engineer* — July 2024 – July 2025
- Reengineered the battery pack of a 3.5-ton electric truck with an improved cell-to-pack ratio by 10%. Carried out the bench testing and cycling of the battery pack.

Speedloop Auto - *Mechanical Design Intern* — January 2023- June 2023
- Performed design calculations for the brake system of a reverse trike vehicle. Successfully attained the braking torque requirement to stop the vehicle on 18% gradient.

Team Octane Racing Electric (COEP) - *Transmission Head and Lead Driver* — July 2022 - July 2023
- Designed and modelled a limited-slip differential with a spring-varied torque bias ratio of 25-40%. Calculated optimum spring stiffness used in the spring pack by simulating on MATLAB. Resulted in improvement of lap time by 1.1s.

ACADEMIC PROJECTS AND SIMULATIONS

Field-Oriented Control of Permanent Magnet Synchronous Motor | MATLAB, Simulink, ANSYS Maxwell
- Simulated differential equation based FOC to evaluate the variance of motor performance with motor parameters.
- Modelled of the motor in ANSYS Maxwell to obtain output parameters.

Modeling of electromagnetic induction machines | SIMULINK
- Energy conversion modeling for linear electromagnetic machine based on differential equations.
- Simulation of 3-phase induction machine using reference frame theory. Testing for electromagnetic power and losses at different load conditions.

Cell Modelling| MATLAB, Simulink
- Formulated a Simulink model of 2-RC equivalent circuit for Li-ion cell to estimate its SoC from Open cell voltage vs SoC curve.

Series-Parallel Hybrid powertrain modeling and supervisory control logic | Simulink, Stateflow
- Sizing of electric motor and internal combustion engine based on Torque-RPM graph and efficiency contours to maximize MPG of the hybrid powertrain in city driving conditions.
- Modeled series-parallel hybrid powertrain architecture in Simulink to compare its efficiency with internal combustion engine and electric powertrain for a particular drive-cycle.
- Designed the supervisory control logic to ensure maximum efficiency considering the battery SoC, vehicle speed, power and torque demand (throttle position) using rule-based approach in Stateflow.

Vehicle Stability Control for Independent Wheel Drive Systems with optimal torque-allocation using MPC
- Developed a hierarchical vehicle stability controller for a four-wheel-independent-drive EV using Mamdani fuzzy logic and Model Predictive Control (MPC) to improve yaw-rate tracking, sideslip control, and torque allocation based on control-oriented state-space model.s
- Implemented the controller enforcing torque, torque-rate, and slip constraints. Validated it via SIL co-simulation in MATLAB/Simulink and CarSim., across double-lane-change and sine-wave steer maneuvers. reducing low-adhesion yaw-rate divergence from ~70 deg/s to ±12 deg/s and keeping sideslip near zero to prevent near spin-out.
- Improved sine-wave steer stability at 100 km/h by reducing peak yaw rate by ~30–45% and peak sideslip by ~75. Reduced low-adhesion yaw-rate divergence from ~70 deg/s to ±12 deg/s and keeping sideslip near zero to prevent near spin-out.

Adaptive Cruise Control and Lane Keep Assist for RC vehicle
- Ultrasonic sensor calibration and noise-filtering using linear Kalman filter to measure the distance between obstacles and vehicle.
- Implemented adaptive cruise control and lane-keep assist using sensor fusion and PID control for throttle and steering to maintain a safe distance from the obstacle in the front as well as on both sides of the RC vehicle.

Engine Dyno Data Analysis, Calibration and Testing
- Calibrated spark timing across load and air-fuel-ratio sweeps on a spark-ignition engine, analyzing heat release rate and CA10/50/90 combustion phasing to identify MBT timing and achieve peak indicated efficiency of 36.6%.
- Optimized diesel injection strategy across start-of-injection, rail-pressure, and speed sweeps, correlating combustion phasing and ignition delay against NOx and CO to map emissions-vs-efficiency tradeoffs for calibration decisions.

- Performed knock analysis on high-load operating points using FFT of in-cylinder pressure, identifying knock frequencies above 5 kHz to inform spark-retard and knock-limit calibration boundaries.
- Quantified cycle-to-cycle combustion stability (COV of IMEP and peak pressure) across operating conditions to define stable calibration limits and lean/dilute operating margins.
- Processed high-speed dynamometer and LabVIEW TDMS test data in MATLAB to compute IMEP, BMEP, FMEP, and brake/indicated fuel conversion efficiency, building the analysis pipeline used to evaluate calibration changes across a 13 L diesel and 2.3 L turbo-gasoline engine.
- Developed and validated a zero-dimensional Wiebe-function combustion model against experimental cylinder pressure, HRR, and mass-fraction-burned data, then ran parametric sweeps on combustion timing, duration, and fuel mass to predict calibration sensitivities ahead of dyno testing.
- Conducted thermodynamic cycle and volumetric-efficiency analysis (valve flow, choked-flow transitions, pumping loops) to understand the airpath and combustion fundamentals underpinning load control and VE-based calibration.

CERTIFICATION AND SKILLS

- Electric Vehicle Design| Great Learning. — September 2023 – October 2024
- Electric Vehicles| Intellipaat, IIT Roorkee — October 2023 – Present

Technical Skills: MATLAB, Simulink, Stateflow, New Eagle Raptor CATIA V5, SolidWorks, Control Systems, Hybrid-powertrain controls, thermodynamics